Math 526, F09
Quiz 12

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Given a matrix $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$.

(a) Find the eigenvalues λ's and corrseponding eigenvectors $\mathbf{x}$'s of $\mathbf{A}$.

$$\det(A-\lambda I) = \det\begin{bmatrix} 1-\lambda & 2 \\ 5 & 4-\lambda \end{bmatrix}$$

$$= (1-\lambda)(4-\lambda) - 2\cdot 5$$

$$= \lambda^2 - 5\lambda - 6$$

$$= (\lambda-6)(\lambda+1) = 0$$

$$\Rightarrow \lambda_1 = 6 \quad \lambda_2 = -1$$

For $\lambda_1 = 6$, solve $(A-6I)x_1 = 0$

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} x_1 = 0 \Rightarrow x_1 = c_1\begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

For $\lambda_2 = -1$, solve $(A+I)x_2 = 0$

$$\begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} x_2 = 0 \Rightarrow x_2 = c_2\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

(b) Find an eigenvalue diagonalization $\mathbf{A} = \mathbf{S}\Lambda\mathbf{S}^{-1}$ based on the results from (a).

Let $c_1 = 1$, $c_2 = 1$, we get $x_1 = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Thus

$$S = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \quad S^{-1} = \frac{1}{(-7)}\begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{7} & \frac{1}{7} \\ \frac{5}{7} & -\frac{2}{7} \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 6 & \\ & -1 \end{bmatrix}$$